



Essay 1: Micro Case Study: Designing a Production Recommender Surface

Due Week 03: Mon, 24 Aug 2026 23:59 SGT

Strict word limit: 1,500 words. The essay body must not exceed this limit. The title, reference list, acknowledgements (including AI use), and any appendices are excluded.

Choose **one** production recommendation surface from the four approved cases below. Focus only on a **single** particular user-facing recommendation surface, rather than the company in general.

Recommender Case	Example recommendation surface
Grab	Grab Feed cards, merchant/food discovery, offers, or cross-service recommendations
Lazada	Product discovery, category/search ranking, campaign product selection, or new-product exposure
Instagram	Explore, Reels, or Feed ranking
LinkedIn	LinkedIn Jobs personalised job recommendations, job alerts, or Hiring Assistant candidate sourcing

For your selected recommendation surface, critically analyse and propose a technically credible recommender design for the particular recommendation aspect you have selected. Draw on the Week 01–02 material (Collaborative Filtering & Latent Factor Models) in constructing and defending your argument. A strong argument accurately applies relevant Week 01–02 recommender concepts and explains why they are appropriate for the chosen recommendation surface.

The structure and emphasis of the essay are deliberately yours to determine. Your task is to make a focused technical argument, rather than to follow a fixed template. You should decide what evidence, explanation, comparison, and level of implementation detail are necessary to persuade an informed reader that the proposed approach is appropriate for the case.

You should not describe a **complete** production recommender system; rather, select one or two technical concepts most consequential to your chosen surface and develop a precise, evidence-based argument.

The case study should be grounded in the selected platform's actual product context, but it is not an exercise in recovering a proprietary algorithm. Clearly distinguish **documented facts** (if any) about the platform from your own **reasoned technical proposal**. The work should engage seriously with algorithmic choice, the realities of operating a recommender over a large and changing catalogue, and the relationship between technical choices and inferred recommendation objectives.

You must also return to the **Week 01 slides** and select one of the card-suit archetypes to use as a substantive analytical lens. The selected lens must feature in your technical design choice, objective, constraint, feature, ranking rule, or evaluation criterion; merely defining the archetype is insufficient.

1 Course Materials, Evidence, and AI Use

Use the Week 01–02 course material and cite **no more than two external sources** in the essay. You should build and consult a wider bibliography during your research, but select the two most useful sources to support the final argument. Cited source should substantiate the platform/surface or support the technical proposal. You need not cite the suggested e-book readings or the course materials; these are implied in your assignment.

Use **APA author–date in-text citations** in the body of the essay — for example, (*Gomez-Uribe & Hunt, 2015*) or *Gomez-Uribe and Hunt (2015) argue that* End the essay with an APA-formatted reference list containing the sources actually cited. Do not treat citations as decorative background: use each source to support a specific factual or technical claim.

Appendices, source notes, and acknowledgements are not part of the essay body and will not be used to assess the technical argument.

2 Assessment and Peer Review

Assessment will focus on the technical soundness and specificity of the argument with respect to the recommendation surface; the disciplined use of evidence; the meaningful application of the Week 01 card-suit lens; and demonstrated ownership of the essay’s ideas and writing. Clear communication and transparent, student-directed AI use are expected.

Essay 1 contributes **10%** to the course assessment.

Component	Weight	Arrangement
Your submitted Essay 1	8%	Assessed on the quality of your individual technical case-study essay.
Peer review	2%	Assess the essay allocated to you by the teaching team, using the provided peer-review process. The peer review will be on a focused aspect of the holistic evaluation done by the instructing staff’s review.

The teaching team will **send each student an allocated essay directly** for peer review. Complete the peer review independently. Do not use generative AI or automated analysis/writing tools to read, interpret, evaluate, draft, revise, or submit the review. Use of approved accessibility support

remains subject to the course and University policy. Peer review will be facilitated through the **Canvas Assignment Component**, which will be made available soon. The peer review period is one week, due on the Week 04 default assignment deadline: **Mon, 31 Aug 2026 23:59 SGT**.

2.1 ⚠ Confidentiality of Assessment Materials

Do **not** post, upload, share, or otherwise disclose your Essay 1, an essay allocated to you for peer review, or your peer-review comments on the public web or through other public-facing channels. These materials are provided for course learning and assessment only. Keeping them non-public preserves the assignment as a viable source of learning for future students.

2.2 ⚠ Limits of Public Technical Disclosure

Public information about production recommender systems is usually thin because key algorithmic, data, serving, and operational details are proprietary. Use the taught course concepts and the permitted sources to make a **clearly labelled technical extrapolation** about how the advertised recommendation surface could be implemented. Do not present an extrapolation as a documented fact about the platform.

2.3 ⚠ Edit Your Essay Carefully!

The 1,500-word limit requires a tightly coherent argument, so make each word and sentence count. An AI will tend to give generalities in answering this assignment, and may do poorly on the assignment. Use AI to support — not substitute for — your own developing understanding of the Week 01–02 material.

3 AI Use, Peer Collaboration, Ownership, and Verification

AI use is encouraged. Use it as a research and thinking partner to identify vocabulary, locate possible lines of inquiry, explain unfamiliar production concepts, generate alternative technical designs, test counterarguments, and improve the clarity of your draft. It is not evidence, however, and all AI-generated claims and citations must be verified against the original sources.

You must **lead the investigation**. Before using AI substantively, establish the scaffolding for its work: identify the recommendation surface, set a provisional question or claim, state the technical issue you wish to investigate, and give the system a focused task. Your prompts should steer the inquiry rather than ask AI to produce an unexamined essay. The central idea, interpretation of the case, technical judgment, organisation, and final argument must be your own. AI may inform your thinking, but it must not replace the ideation that gives the essay its intellectual direction.

In an **Acknowledgements** section after the Reference List, attach an **AI-use declaration**. It must identify the tool(s) used; explain how you framed, constrained, or scaffolded the AI's investigation; state how AI contributed; and give two examples of outputs that you checked, corrected, or rejected. Retain your relevant prompts and outputs in case they are requested.

Students are encouraged to discuss the general aspects concerning your recommender case to further your understanding of the course's topics. Discussion may clarify concepts and test reasoning, but students must not co-author prose or submit materially shared technical arguments. Discussions that substantially affected your architectural decisions should also be declared in your **Acknowledgements** section. Note: the permission to discuss general course concepts does not apply to an allocated peer essay or to the preparation of your peer review.

Students may be asked to discuss or explain their essay, its sources, its technical choices, and their use of AI. This is an opportunity to demonstrate ownership of the work. The instructor's academic judgment, including the determination of authorship, the assessment of the work, and the final grade, is final.

4 Seed References

You may consult any number of sources while researching, but your essay may cite **no more than two external sources** in APA author–date style. Treat practitioner, vendor, and social-media sources with appropriate caution and corroborate them before you make factual claims.

- Grab
 - [Making Grab's Everyday App Super](#)
 - [How Grab Is Accelerating Growth with Real-Time Personalization Using CDP Scenarios](#)
- Lazada
 - [Hierarchical Interest Modeling of Long-tailed Users for Click-Through Rate Prediction](#)
 - [Improved Accuracy, Declining Orders: Uncovering the Vicious Competition Trap in Multi-Domain Recommendation](#)
- Instagram
 - [Designing a Constrained Exploration System](#)
 - [How Instagram Feed Recommendations Work](#)
 - [Scaling the Instagram Explore Recommendation System](#)
- LinkedIn
 - [Learning to Retrieve for Job Matching](#)
 - [How We Engineered LinkedIn's Hiring Assistant](#)
 - [Improving Job Matching with Machine-Learned Activity Features](#)